

Problem Statement

This capstone project works at the intersection of advanced technology and healthcare, improving ASD diagnosis accuracy by incorporating VQA into ASD diagnosis.

The VQA system is embedded in a pipeline that generates video answers and predicts the likelihood of Autism Spectrum Disorder (ASD).

Background

Current methods of Autism diagnosis such as interviews, tools, evaluations, imaging, and questionnaires, are time-consuming and costly and also lack a comprehensive approach.

Using images and videos, especially facial expression analysis, provides a comprehensive and effective method for ASD detection, contributing to the development of more supportive strategies.

Dataset and Features /Project Requirements

Video datasets of ASD and TD children (aged 5 to 7 years) were taken from YouTube and recorded from clinical sources—12 videos from BMCRI, ASHA, and Udipi Government School.

- Facial expression features are extracted using OpenFace and Py-Feat.
- 11 question categories with close-ended answers (yes/no) and answer generation with three threshold types (manual, mean and mode).



Fig 1. Dataset of ASD and TD

Design Approach / Methods

The unique bi-modal model integrates dual LSTM layers for text processing and facial expression features including various facial Action Units (AUs). The first layer captures sequential information, while the second refines for longer-term dependencies, integrating facial features for a fused vector.

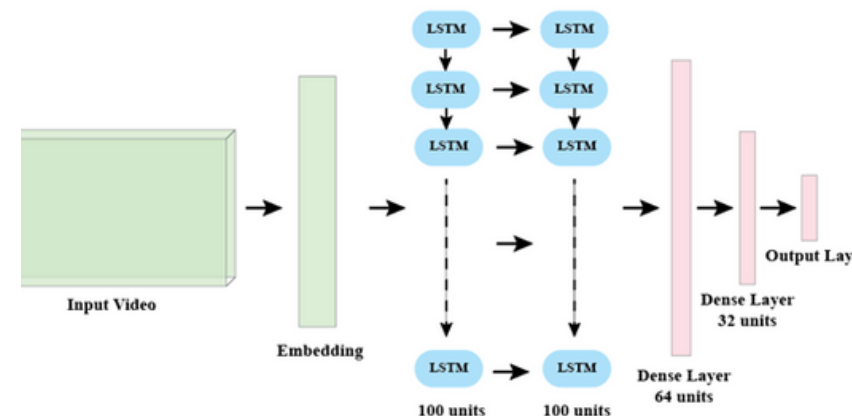


Fig 2. Architecture of VQAutism

Results and Discussion

Model 1 : BERT-based transformer model

The computational complexity and resource demands of this model are high.

Model 2 : Artificial Neural Network(ANN)

The simpler ANN performs well in resource-limited scenarios, but with compromises in contextual understanding.

Model 3 ~ our proposed VQAutism (98% accuracy)

VQAutism emerges as the top-performing model. Its efficacy lies in integrating textual and facial information build on top of LSTM layers

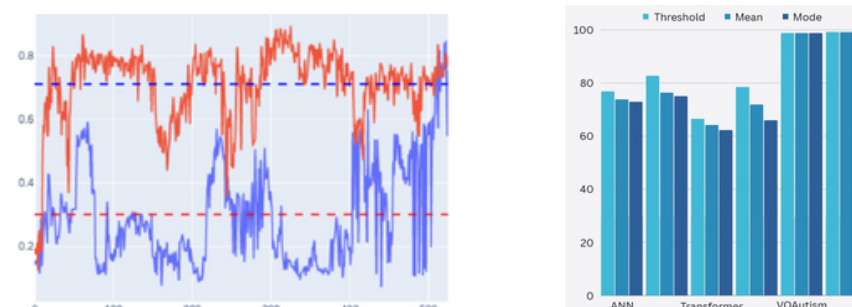


Fig 3. Features of ASD and TD Fig 4. Accuracy for ANN, Transformers and VQAutism on Youtube and Clinical

Summary of Project Outcome

Beyond its initial scope, VQAutism emerges as a powerful tool with wide-ranging applications, promising advancements in video-based diagnostics across diverse healthcare domains.

The alignment of the outcomes with the initial estimates provides strong validation for the project's objectives and methodologies. Various facial expression elements increase emotion comprehension. The model's success in effectively unifying textual and facial information exceeded expectations, establishing VQAutism as a promising video classification model with 98% accuracy.



Fig 5. Project Pipeline

Conclusions and Future Work

PVQAutism emerges as a powerful tool with broad applications, and promising advancements in video-based diagnostics across diverse healthcare domains. Introducing data augmentation, transfer learning, multimodal data fusion, ethical issues, long-term research, community engagement, impact evaluation, and comparison with current tools are all key areas for future exploration.

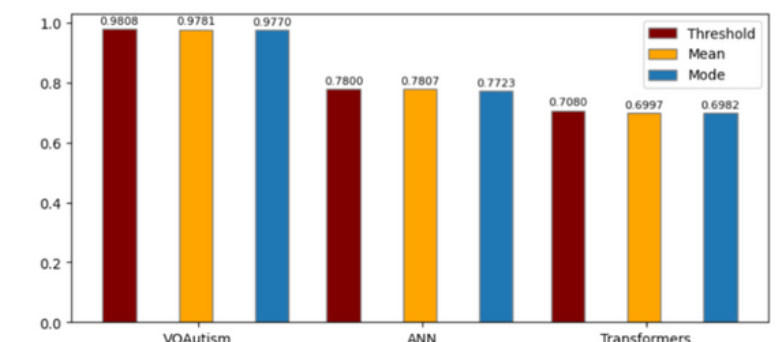


Fig 6. AUC-ROC Scores for models

References

[1] Tariq, Q., Daniels, J., Schwartz, J. N., Washington, P., Kalantarian, H., & Wall, D. P. (2018). Mobile detection of autism through machine learning on home video: A development and prospective validation study. PLoS medicine, 15(11), e1002705.